



Koneru Lakshmaiah Education Foundation

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DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT (25CS1302E)

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 5	Team No- !	Project Title- JARVIS - Smart Personal Assistant
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1. Abstract

JARVIS is a locally-hosted, AI-powered personal assistant built entirely in Python, designed to operate as a full-stack intelligent system combining a structured relational backend, distributed LLM routing, semantic memory, voice interaction, and complete computer control.

At its core, the system implements a layered backend architecture: HTTP requests arrive at a FastAPI service layer, pass through Pydantic validation, are dispatched by a multi-agent orchestration engine (LangChain + LangGraph), and are resolved against a dual-store memory system — a normalized SQLite database (facts, preferences, entities, reminders, conversation episodes — designed to 3NF using SQLAlchemy ORM) and a ChromaDB vector store for semantic similarity retrieval with GPU-accelerated embeddings.

The backend applies all core database engineering principles: DDL/DML via SQLAlchemy, B-tree indexing on high-frequency query columns, ACID-compliant transaction boundaries with rollback support, and concurrency-safe multi-threaded writes using threading locks. The distributed intelligence layer implements multi-provider LLM rotation (Ollama → Google Gemini multi-key → Anthropic Claude → OpenAI), with quota-aware cooldown tracking and automatic local fallback on network partition — demonstrating CAP theorem trade-offs in a real production system.

The system additionally supports multilingual voice I/O (English + Telugu), 61 computer-control tools (mouse, keyboard, shell, browser, files, power), and a glassmorphism desktop HUD with live system stats — all unified through a single conversational interface. JARVIS demonstrates the end-to-end application of database systems engineering and distributed backend principles in an intelligent, real-world autonomous system.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520030369	Manne Yashwanth Manoj	
2520030383	Kheerthi Ganta	

3. Frameworks / Platforms / Software Used

(List the frameworks, libraries, platforms, or tools you plan to use.)

Category	Tools / Libraries
Language	Python 3.11+
Database (Relational)	SQLite (via SQLAlchemy ORM 2.0 + Alembic migrations)
Database (Vector/Semantic)	ChromaDB 0.5+ with SentenceTransformer GPU embeddings
API Backend	FastAPI 0.115+, Uvicorn, WebSockets, Pydantic v2
Agent Orchestration	LangChain 0.3+, LangGraph 0.2+, custom multi-agent CEA
Local LLM	Ollama (phi3.5:3.8b, gemma2:2b, qwen2.5-coder:3b)
Cloud LLM	Google Gemini (multi-key rotation), Anthropic Claude, OpenAI GPT-4o
Voice STT	sounddevice + Google Speech Recognition + faster-whisper
Voice TTS	Microsoft Edge Neural TTS (en-GB-RyanNeural), gTTS (Telugu)
Computer Control	PyAutoGUI, pywin32, psutil, Playwright, mss, EasyOCR
Desktop App	pywebview (WebView2/Edge runtime), HTML/CSS glassmorphism UI
Async & Concurrency	asyncio, threading.Lock, aiofiles, asyncio-throttle
Logging & CLI	Loguru, Rich, Click, Colorama
Testing	pytest, pytest-asyncio, pytest-cov
Configuration	python-dotenv, PyYAML, pydantic-settings
OS	Windows 11 (hardware: RTX 4050 6GB, 16GB RAM)

Guide Name and Signature (remarks if any) _____